A. Project Understanding and Scope

The assigned final project is the Universe 25 Population Simulator, a Python desktop application that simulates changes in a mouse population across multiple generations. Its purpose is to explore how the initial male and female populations, reproduction rates, death rates, and limited resources affect population growth and survival. The application aims to solve the problem of understanding population changes through a simplified, assumption-driven model that allows users to adjust parameters and observe the results.

The mandatory features include scenario setup for initial male and female populations, starting resources, and the number of generations; configurable birth rates, baseline death rates, and resource replenishment or carrying limits; a generation-based model that calculates births, deaths, resource consumption, and replenishment; resource constraints that prevent impossible reproduction and negative population or resource values; Run, Step, and Reset controls; and generation-by-generation reports and charts showing population and resource changes. The application must also identify the final state of the simulation and validate inputs to prevent invalid settings from causing errors.

The expected inputs are the initial numbers of male and female mice, starting resources, the number of generations from 1 to 100, birth and death rates, and resource replenishment settings. The processing operations involve calculating births and deaths, updating the population and resources for each generation, applying resource limitations, and checking that all values remain valid. The expected outputs are a generation-by-generation results table, population and resource charts, and a final summary of the simulation. The application must maintain the rules that the next population equals the current population plus births minus deaths, and that the male and female counts add up to the total population.

The project is limited to a simplified, generation-based population model and is not intended to scientifically reproduce the historical Universe 25 experiment. It does not include individual mouse agents, detailed social behaviors, or claims about human society. The results depend on the assumptions and rules implemented in the program, so they should be interpreted as simulated outcomes rather than accurate predictions of real-world population behavior.